Cognitive Trade-offs Between Speech Disfluencies and Active Engagement from Proactive AI-Mediated Scaffolds in Turn-taking

Overview. We tested how proactive AI-mediation in turn-taking to scaffold L2 speakers would affect their speech flow and perceived engagement. We found that proactive scaffolds do not merely reduce disfluency but reorganize it into different processing signals. While less intrusive mediation improves efficiency and engagement, it imposes a cognitive tax—specifically in the evaluation of more intrusion scaffolds—that manifests as distinct disfluency patterns.

Introduction. Turn-taking is one of the most cognitively demanding processes in human communication, requiring speakers to simultaneously comprehend input, predict content, and formulate responses. For L2, prediction errors at these bottlenecks often manifest as speech disfluencies (e.g., silent pauses, repairs), which can deteriorate engagement and alignment. While existing assistive technologies are predominantly reactive, we propose that *proactive* AI-mediated scaffolding can redistribute this cognitive load, though it may introduce new attentional costs. This study evaluates a proactive system designed to scaffold turn-taking via three features targeting specific processing bottlenecks: *lexical clarifications* (comprehension), *idea suggestions* (production), and *topic summaries* (working memory). We investigated how these proactive mediations shape the frequency and timing of disfluency markers and their relationship to perceived engagement.

Method. We conducted a laboratory-based Wizard-of-Oz study with 35 L2 English speakers. Participants engaged in a standardized 15-minute collaborative task (designing a fusion restaurant menu) with a confederate while interacting with the proactive tool embedded in Zoom meeting. The system proactively provided context-sensitive clarifications for jargon/cultural terms, content suggestions for idea generation, and summaries at topic transitions from the confederate's speech. We analyzed speech disfluency patterns (silent pauses, filled pauses, repairs, hesitations) via recordings of the user study sessions and aggregated the findings from the qualitative feedback via follow-up interviews (N=29). Disfluency types were manually coded, serving as diagnostic indicators of processing demand, and mixed-effects models were used to predict the relationship between feature usage, disfluency rates, and perceived engagement.

Results and discussion. Results revealed a critical cognitive trade-off between perceived engagement and speech fluency. **(1) Disfluency Mechanics:** Feature type significantly modulated disfluency patterns. *Clarifications* significantly reduced uncertainty-related delays by resolving comprehension gaps immediately. Conversely, *suggestion* turns yielded a higher incidence of disfluencies, characterized by frequent but shorter micro-disruptions. This reflects the extra cognitive cost of the simultaneous evaluation and production (a dual-task conflict) from the real-time AI-mediation. **(2) The Production-Disfluency Paradox:** Despite the increased objective disfluency load associated with suggestions, participants perceived the conversation as smoother and themselves as more engaged. High-proficiency users and critical evaluators (identified via Latent Class Analysis) exhibited distinct trade-offs: increased *silent pauses* significantly predicted lower overall ratings whereas *uncertainty/delay* markers did not negatively impact satisfaction, likely because they signaled active integration of assistance. **(3) Autonomy Gradient:** There was a clear preference for autonomy-preserving scaffolds. *Summaries* and *clarifications* (less intrusive) received higher ratings, while *suggestions* (more intrusive) created tension between cognitive benefit and attentional cost. This work extends psycholinguistic theories of turn-taking to AI-mediated contexts, showing that optimal AI scaffolding must balance this trade-off by prioritizing concise, autonomy-preserving interventions and carefully timing high-inference suggestions to align with human predictive windows.

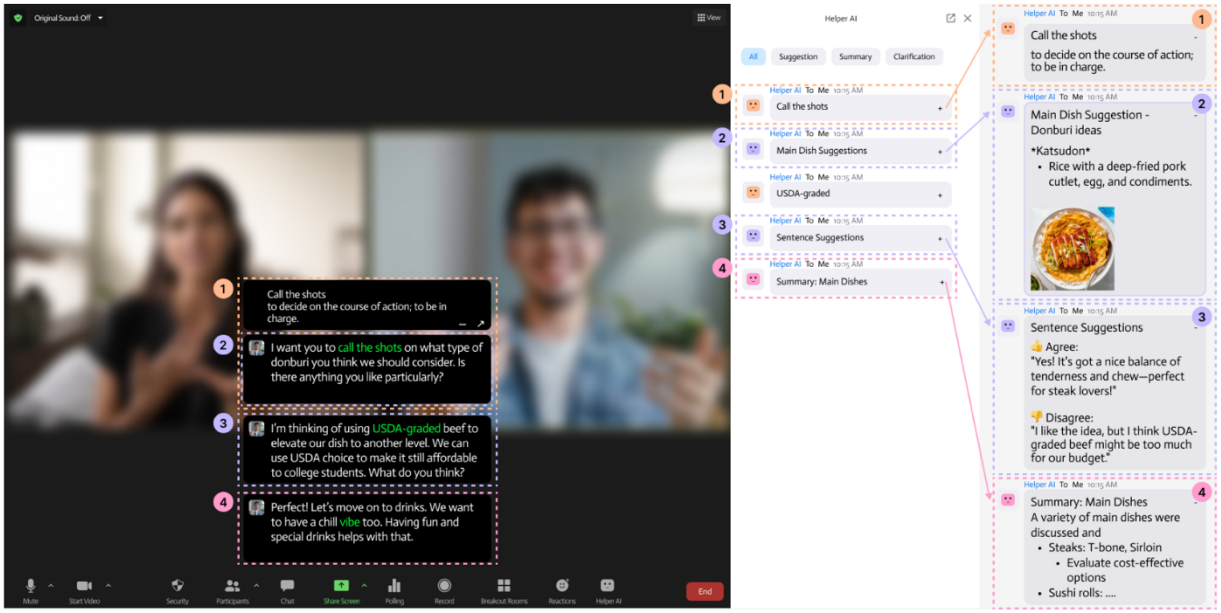

Figure 1: Zoom interface with XPLAIN; 1 : clarification box popped up above caption once clicked with minimal eye movement track, saved as minimized in the side bar for future reference; 2 : idea suggestion; 3 : full-sentence suggestion; 4 : summary. 2 3 4 all popped up as expanded windows. Only one feature expands at a time to avoid visual overload. (N.B. the caption stacks were for illustration purposes: only one caption box stayed at the bottom position as the conversation progressed.)

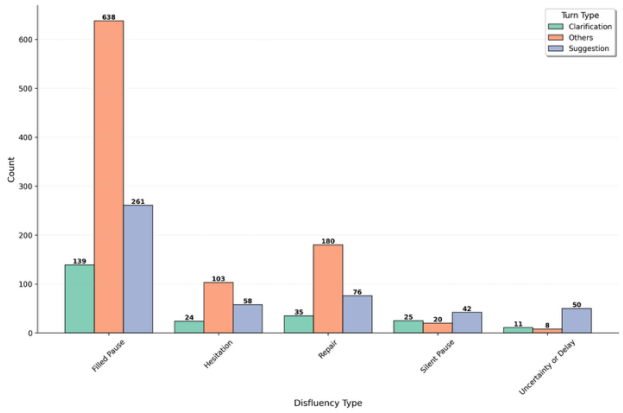

Figure 2: Disfluency Count by Turn Type and Disfluency Type

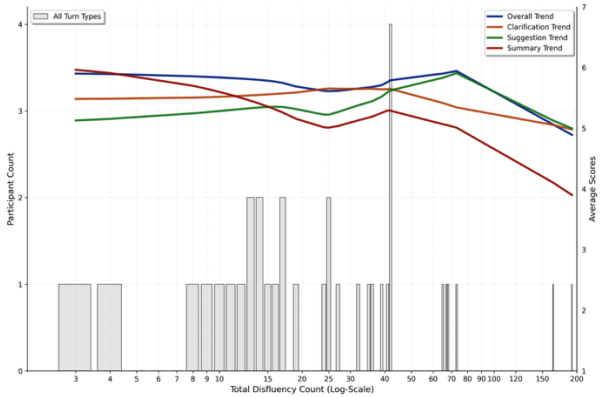

Figure 3: Distribution of Total Disfluency Count and Feature-Specific Rating